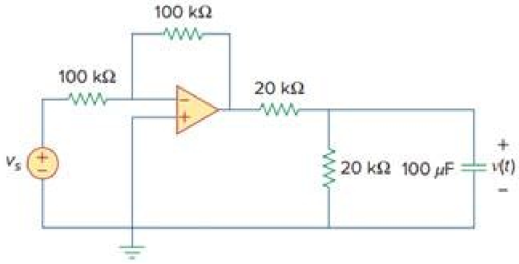
Problem

For the op amp circuit in Fig. 7.136, suppose *vs* = 10*u*(*t*) V. Find *v*(*t*) for *t* > 0.

Fig. 7.136:



Step-by-step solution

Step 1 of 7

Refer to Figure 7.136 in the textbook.

Draw the circuit diagram with notations.

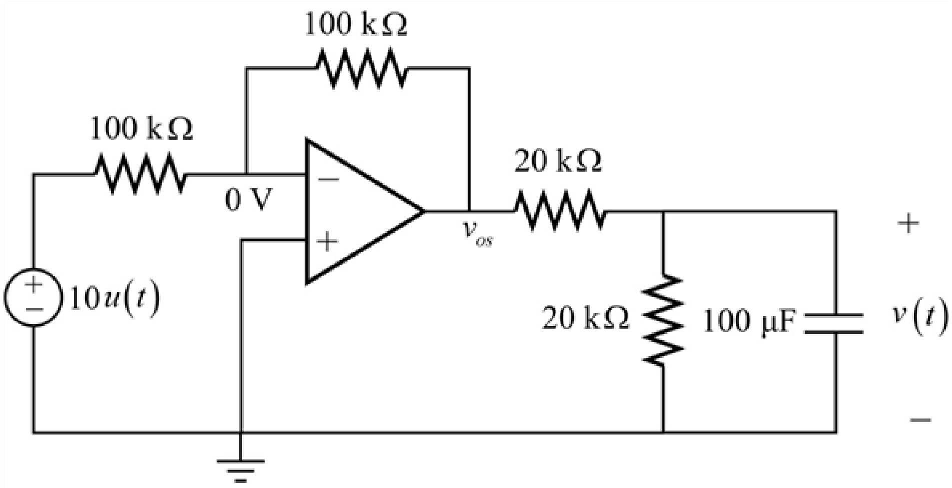


Figure 1

Step 2 of 7

As per virtual ground concept, the voltage at the inverting terminal is equal the voltage at the non-inverting terminal.

$$v_p = v_n = 0 \text{ V}$$

Apply Kirchhoff's current law at the inverting terminal.

$$\frac{v_n - 10u(t)}{100 \text{ k}\Omega} + \frac{v_n - v_{os}}{100 \text{ k}\Omega} = 0$$

$$\frac{0 - 10u(t)}{100 \text{ k}\Omega} + \frac{0 - v_{os}}{100 \text{ k}\Omega} = 0$$

$$-10u(t) - v_{os} = 0$$

$$v_{os} = -10u(t) \text{ V}$$

Step 3 of 7

Replace the output voltage of the op-amp with the voltage source.

Draw the modified circuit diagram.

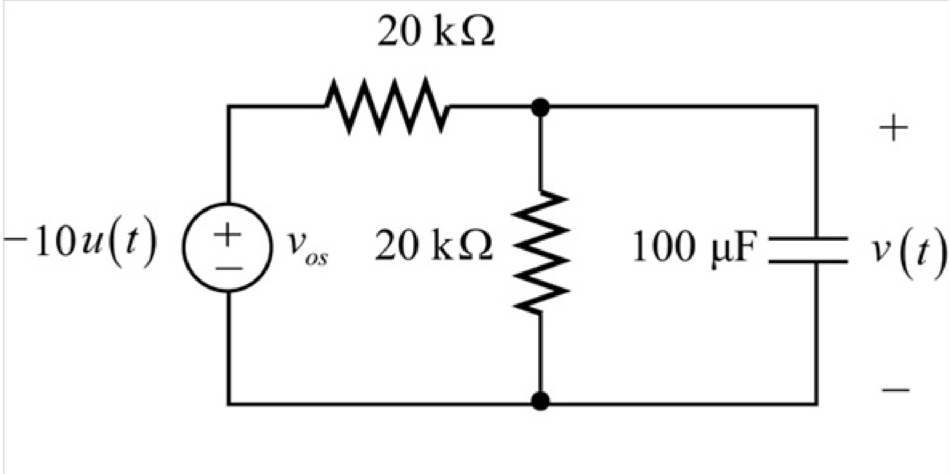


Figure 2

Step 4 of 7

Consider the properties of step input.

$$u(t)=\begin{cases}0 & t<0\\ 1 & t>0\end{cases}$$

The circuit is source free for $t<0$, so the initial voltage across the capacitor is zero.

$v(0)=0\text{ V}$

Draw the circuit to calculate the final voltage across the capacitor.

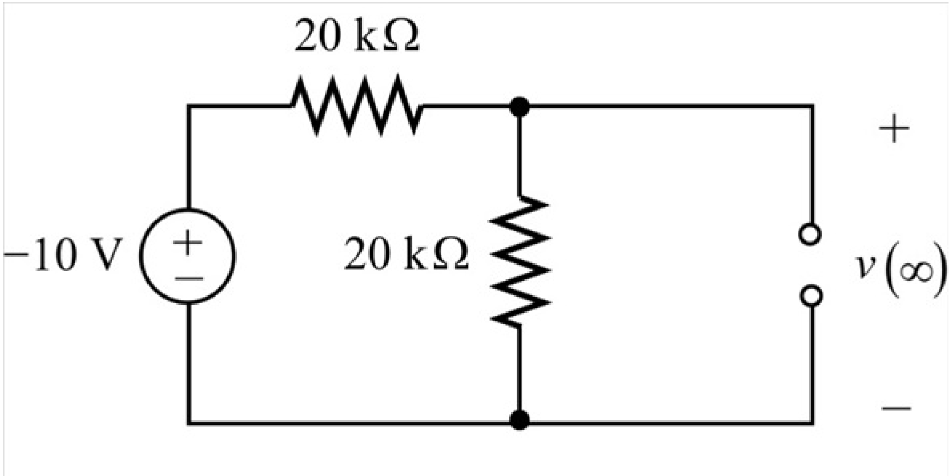


Figure 3

Step 5 of 7

Apply voltage division rule to calculate the final capacitor voltage.

$$v(\infty) = \left(\frac{20\ \text{k}\Omega}{20\ \text{k}\Omega + 20\ \text{k}\Omega}\right)(-10)$$
$$= \left(\frac{1}{2}\right)(-10)$$
$$= -5\ \text{V}$$

Draw the circuit to calculate the equivalent resistance for $t > 0$.

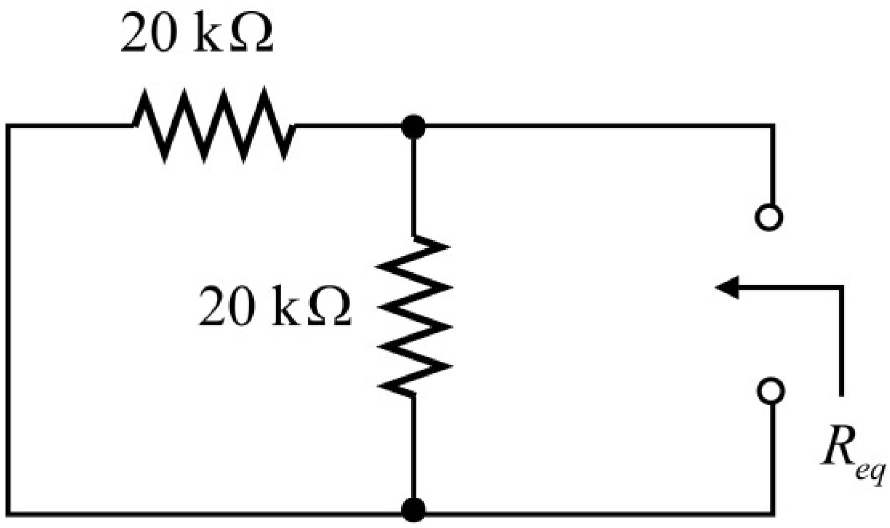


Figure 4

Step 6 of 7

The equivalent resistance for $t > 0$ is,

$$R_{eq} = 20\ \text{k}\Omega \,||\, 20\ \text{k}\Omega$$
$$= \frac{20 \times 20}{20 + 20}\ \text{k}\Omega$$
$$= 10\ \text{k}\Omega$$

Calculate the value of time constant, τ.

$$\tau = R_{eq}C$$

Substitute $20\ \text{k}\Omega$ for R_{eq} and $100\ \mu\text{F}$ for C.

$$\tau = (10 \times 10^3)(100 \times 10^{-6})$$
$$= 1\ \text{sec}$$

Step 7 of 7

Calculate the expression for the capacitor voltage, $v(t)$.

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}$$

Substitute $0\ \text{V}$ for $v(0)$, $-5\ \text{V}$ for $v(\infty)$ and $1\ \text{sec}$ for τ.

$$v(t) = -5 + (0 + 5)e^{-\frac{t}{1}}$$
$$= (-5 + 5e^{-t})u(t)\,\text{V}$$

Thus, the expression for the capacitor voltage, $v(t)$ is $\boxed{(-5 + 5e^{-t})u(t)\,\text{V}}$.